Atom Interferometry with Bose-Einstein Condensate

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Bose-Einstein condensate(BEC) is a novel matter state with favorable coherence properties. It therefore has great potential in precision measurement and has been used in atom interferometry. In this paper, we use Gross-Pitaevskii equation (GPE) to model a BEC. By reducing the GPE to 1-D, our simulation is efficient enough to reach spatial and temporal resolution that enable the simulation of a BEC interferometer. It's based on contrast interferometry referring to an experimental work [1] by Gupta et al. In our simulation, the measurement of Planck's constant h achieved a measurement uncertainty comparable to simulation accuracy and we observed a systematic error of 1.4×10^{-4} which is similar to that reported in the initial experiment. We propose that this error originated from the beam splitting process and verified this proposition with our simulation.

I. INTRODUCTION

Interferometry has always been a significant promotion to new discoveries of physics. The introduction of laser into interferometry provided sources with better time coherence, which made interferometry-based precision measurement like LIGO (Laser Interferometer Gravitational-Wave Observatory) possible.

Interferometry in essence is an application of wave's diffraction property, and the resolution of an interferometer is approximately proportional to the wavelength. In 1924, de Brogile proposed that every particle should exhibit wavelike behavior[2], and the typical de Brogile wavelength of an atom at room temperature has a magnitude of 1nm, far smaller than that of light, thus having greater utility in various fields, such as the measurement of gravitational constant G, detection of rotation, and test of atomic physics, e.g. the mearsurement of fine structure constant α. In 1985, Chebotayev et al. made the first atom interferometer with thermal atom beam[3], which had a good signal-to-noise ratio but a broad longitude momentum distribution, lowering its sensitivity.

The introduction of Bose-Einstein condensate(BEC) solves the problem. BEC was first predicted by Bose[4] and Einstein[5] in 1924, and was finally produced in the laboratory in 1995 with sodium atoms[6]. It has a narrow momentum distribution and correspondingly long coherent length, which greatly promoted the development of atom interferometry as laser did to traditional light interferometry. M. R. Andrews et al. observed the interference fringes between two bunch of BECs[7], and one of the earliest BEC interferometries was achieved by Gupta et al. in 2002[1].

Although there have been quite a lot of numerical simulation methods of BECs [8][9][10][11] since the observation of BEC, the discussions on the simulation of BEC interferometers remains few. The temporal and spatial extents of the interferometer are challenging in terms of computation resources [12]. A scaling method and an approximation of the free propagation were introduced in Ref. [12] therefore. In this paper, we discuss the fundamental aspects of BEC interferometry and choose a 1-D approximation [8]. We show our method enables the realization of numerical simulation of a contrast interferometer and a simulated measurement of h. Some comparisons between simulations and theoretical results are also made to validate our methods.

We will start with discussions on the essential equations that models a BEC in Sec. II A. Then in Sec. III, we will introduce some simulation algorithms and approximations which enable our numerical simulation of a BEC. A simple demonstration of BEC interference is made after that. Sec. IV includes simulation techniques involving atom scattering and the extraction of contrast signals of the interference. We then realize a virtual BEC interferometer based on a contrast interferometry experiment [1]. In Sec. V, we present our simulation results. A measurement precision comparable to simulation accuracy is realized, but a systematic bias of 1.4×10^{-4} is observed as is in the original experiment. We proposed an explanation and verified it with our simulation.

II. MODELING BEC: GROSS-PITAEVSKII EQUATION

A. Wave function discription and order parameter

Particles that obey the Bose-Einstein distribution

$$n(\epsilon_i) = \frac{1}{e^{\beta(\epsilon_i - \mu)} - 1}, \beta = \frac{1}{k_B T} \tag{1}$$

are called bosons. Let us consider a Boson system described in terms of a many body wave function $\Psi_N(\boldsymbol{r}_1, \boldsymbol{r}_2, ...\boldsymbol{r}_N)$. It is derived by statistical mechanics that when the system is cooled under a critical temperature it undergoes a phase transition and macroscopic atoms are transferred to the ground state, denoted by $\phi_0(\boldsymbol{r})$. When all the N particles are in the state of the lowest energy, and consider the commutative symmetry of Boson system, then the many body wave function takes

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the form

$$\Psi_N(\boldsymbol{r}_1, \boldsymbol{r}_2, ...\boldsymbol{r}_N) = \prod_{i=1}^{N} \phi_0(\boldsymbol{r}_i) \tag{2}$$

which represents the ground state of the system. As a result, we can define the order parameter simply by the single-particle ground state with normalization to the occupation number as[11]

$$\psi(\boldsymbol{r}) = \sqrt{N_0}\phi_0(\boldsymbol{r}) \tag{3}$$

with the density of the condensate taking the form $n(\boldsymbol{r}) = \|\psi(\boldsymbol{r})\|^2$.

B. Full Gross-Pitaevskii equation

In this paper we consider the nonideal gases so the interaction between atoms cannot be neglected. As a result, we expect the governing equation has a nonlinear form taking into account the interaction with the mean field produced by other bosons. According to [11], the interaction energy is given by $E_{int} = \frac{g}{2}\int \|\psi(\boldsymbol{r})\|^4 d\boldsymbol{r}$ where $g = \frac{4\pi\hbar^2 a_s}{m}$, and the total energy given by

$$E = \int \left(\frac{\hbar^2}{2m}\|\nabla\psi\|^2 + V\|\psi\|^2 + \frac{g}{2}\|\psi\|^4 \right) d\boldsymbol{r} \tag{4}$$

is a functional of the order parameter ψ. By using the variational ansatz $i\hbar\frac{\partial\psi}{\partial t} = \frac{\delta E}{\delta\psi^*}$, we get the 3-dimensional full Gross-Pitaevskii equation (3-D GPE) as

$$i\hbar\frac{\partial\psi}{\partial t} = \left(-\frac{\hbar^2}{2m}\Delta + V + g\|\psi\|^2 \right)\psi \tag{5}$$

which describes the evolution of BEC in an external potential V.

C. From 3-D GPE to 1-D GPE

Due to computational limitations, we encountered difficulties in solving 3-D GPE with spatial resolution and evolution time that are high enough for simulation of an interferometer. An approximation of 3-D GPE by dimension reduction is therefore of significance.

In our discussions, the BECs used in interferometry are first trapped in a 3-D harmonic potential. When the trap is turned off in one of the three directions (denoted as x), the BECs are scattered, propagate and interfere in a "waveguide" along the x axis.

Therefore, external potentials can be considered to possess cylindrical symmetry about x axis, in particular those with the form of $U(\boldsymbol{r}) = \frac{1}{2}m\omega_\perp^2(y^2+z^2) + V(x)$. So an ansatz to the 3-D GPE takes the form of Gaussian function along the transverse (y and z) directions [8]:

$$\psi(\boldsymbol{r}, t) = \frac{e^{-(y^2+z^2)/2\sigma(x,t)^2}}{\pi^{1/2}\sigma(x,t)} f(x,t). \tag{6}$$

Using the slow varying approximation in the axial direction ($\nabla^2 \approx \nabla_y^2 + \nabla_z^2$) and substituting Eq. 6 into Eq. 5, we get the reduced 1-D GPE [8]:

$$\begin{aligned} i\hbar\frac{\partial f}{\partial t} = &\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V + \frac{gN}{2\pi a_\perp^2}\frac{\|f\|^2}{\sqrt{1+2a_s\|f\|^2}} \right) f \\ &+ \frac{\hbar\omega_\perp}{2}\left(\frac{1}{\sqrt{1+2a_s\|f\|^2}} + \sqrt{1+2a_s\|f\|^2} \right) f. \end{aligned} \tag{7}$$

, where g is $\frac{4\pi\hbar^2 a_s}{m}$ as given above, and $a_\perp = \sqrt{\frac{\hbar}{m\omega_\perp}}$ characterizes the transverse width. The terms containing $a_\perp$ and $\omega_\perp$ represent the influence of the density distribution of in the transverse (y and z) directions, reduced into the mean field. It is worth mentioning that the x axis distribution $\|f(x,t)\|^2$ is defined to be already integrated in the y and z directions.

III. NUMERICAL METHODS

Having completed the formulation of GPE, we are, in principle, enabled to simulate the evolution of BECs. However, numerical methods are yet to be developed. In this section we will introduce the methods we use to solve the 1-D GPE and to get the initial state. To demonstrate them, some results relevant to the first BEC interference experiment in 1997[7] will be given. We will stick to these methods in the subsequent sections in more complex scenarios.

A. Split Step Fourier Method

The GPE, even reduced to 1-dimensional, is still a nonlinear PDE difficult to solve. We implement split step Fourier method [9] to solve it numerically. Consider a 1-D GPE: $\frac{\partial}{\partial t}\psi = H(x,t,\|\psi\|)\psi$, or

$$\psi(x, t+dt) = e^{-\frac{idt}{\hbar}\left(\frac{\hbar^2\partial^2}{2m\partial x^2} + V_{ext} + f(\|\psi\|)\right)}\psi(x,t) \tag{8}$$

To deal with the derivative against x, we intend to separate the $e^{-\frac{idt}{\hbar}\frac{\hbar^2\partial^2}{2m\partial x^2}}$ term. By carefully choosing the form of separation, we can reach an accuracy of O($(dt)^3$) [13] by using

$$\psi(x, t+dt) \approx e^{-\frac{idt}{\hbar}\frac{\hbar^2\partial^2}{4m\partial x^2}} e^{-\frac{idt}{\hbar}(V_{ext}+f(\|\psi\|))} e^{-\frac{idt}{\hbar}\frac{\hbar^2\partial^2}{4m\partial x^2}}\psi(x,t) \tag{9}$$

After separation, the 3 exponents can be regarded as 3 sequential operators performed on $\psi(x,t)$. The 1st and 3rd can be calculated by FFT where $\frac{\partial^2}{\partial x^2}$ is replaced by $\times k^2$ in frequency domain and the 2nd operator is a simple multiplication.

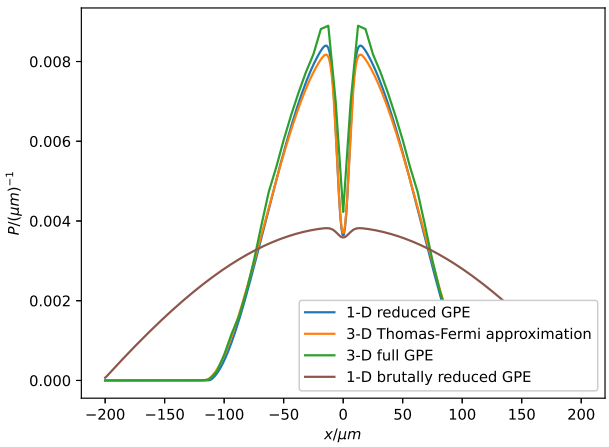

FIG. 1. The BEC contains 5×10^6 Na atoms and is confined in a potential trap with frequencies $(19, 250, 250)\times 2\pi$ Hz[7][14] and a Gaussian splitting potential depending only on x [14]. The density distribution about x (having been integrated along y and z axes) is plotted. The 1-D solution lies close to full GPE, thus, the former is verified to be a good approximation. An analytical solution given with Thomas-Fermi approximation is also plotted. No significant bias in the results is observed, validating again our results. The 3-D GPE has a resolution notably lower because of our computational constraints.

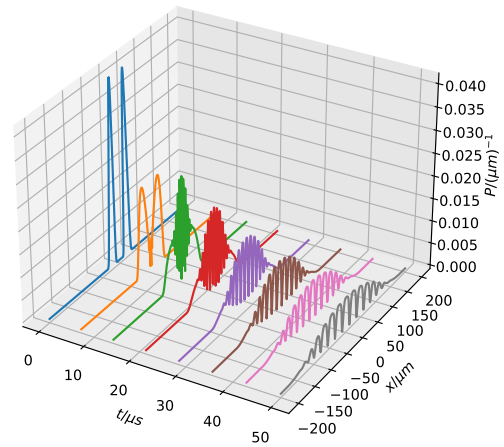

FIG. 2. The evolution of two BECs' number density distribution and their interference is demonstrated. Due to limits of our simulation resolution, the particle number is decreased to 5×10^4. It can be seen from the graph that the BEC is fully separated in the initial state and the interference fringes periods are increasing by time.

B. Simulation of the Ground State

The initial state of a BEC we study is condensed in the ground state. In order to find the wave function giving the lowest energy, we express the total energy in terms of ψ (Eq. 4). The energy is minimized using gradient descent method combined with normalization of ψ every step. The descent direction is given by

$$\psi_{i+1}^{temp} - \psi_i = -\frac{\hbar^2}{2m}\Delta\psi_i + V\psi_i + g\|\psi_i\|^2\psi_i \tag{10}$$

and normalization

$$\psi_{i+1} = \psi_{i+1}^{temp}\sqrt{\frac{N}{\|\int \psi_{i+1}^{temp}\psi_{i+1}^{*temp}d\vec{r}\|}} \tag{11}$$

Eq. 10 is almost identical to GPE (Eq. 5), except for the time t substituted with imaginary time it. This suggests the interpretation of gradient descent as an exponential decay of all possible states [13] other than the state of the lowest energy, i.e. the ground state.

The choice of descent step length is identical to that in numerical solution of GPE. We demonstrated the calculation of the ground state and the parameters are set referring to [7], as is shown in Fig. 1. A comparison among 1-D GPE (Eq. 7), analytical solution using Thomas-Fermi approximation of 3-D GPE (Appendix A) and full 3-D GPE (Eq. 5) are made to show the validity of our chosen 1-D GPE.

A simple but often used [8][9][15] 1-D GPE shows large deviation in Fig. 1 (brown line), in which the effect of density distribution along x axis on the transverse width of Gaussian wave packet is not considered, i.e. the correction terms in Eq. 7, $\sqrt{1+2a_s\|f\|^2}$, are set to 1. It can be seen that our 1-D reduced GPE (Eq. 7, or blue line in Fig. 1) significantly outperforms this simple one in approximating 3-D GPEs and will therefore be used later.

C. Demonstration of BEC Interference

After developing the numerical methods to simulate the ground states and propagation of BECs, we are able to simulate BEC interference. As both a demonstration and a validation, we are to simulate the first BEC interference experiment [7] in this subsection.

The external potential is similar to the last subsection. A harmonic potential traps the BEC [16], while a Gaussian shaped potential along x axis is used to model the laser beam focused into the center of the harmonic trap. A double well trap is thus formed and the BEC can be coherently separated [17] into two "cigar shaped" parts.

When the trap in the x direction and the laser are turned off, two separated BECs will expand and exhibit interference properties when they overlap spatially. An illustrative explanation of the interference fringes can be given by considering two separated symmetric wave pack-

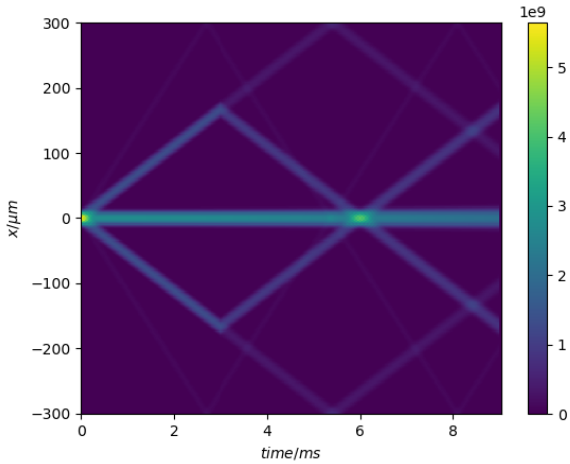

FIG. 3. The atom density (represented by color) is plotted along the time axis. Before $t = 0$, the atoms are trapped in a harmonic potential with frequencies $(20, 50, 50) \times 2\pi$ Hz with zero momentum$|g, 0\hbar k\rangle$. At $t = 0$ it is coherently split by a Kapitza-Dirac pulse mostly into the $|g, \pm 2\hbar k\rangle$ and $|g, 0\hbar k\rangle$ momentum states. At $t = T = 3$ms, the two paths corresponding to the first order scattering are reflected by a 2nd-Brag-π-pulse whose interaction time in Eq. 16 satisfies $\tau = \frac{\pi}{\omega_R^{(4)}}$. At $t = 2T = 6$ms the three paths recombine and we can extract the contrast signal by a probe light beam.

ets $f(x - x_0)$ and $f(x + x_0)$.

Perform Fourier transform on them to get

$$\begin{aligned}\mathcal{F}f(x - x_0) + \mathcal{F}f(x + x_0) &= (e^{ikx_0} + e^{-ikx_0})\mathcal{F}f(x) \\ &= 2\cos(kx_0)\mathcal{F}f(x)\end{aligned} \tag{12}$$

Although the wave packets are not overlapped in space, they overlap and interfere in the frequency domain. Substituting $p = \hbar k$ with $m\frac{x}{t}$ yields the time dependence of interference fringes period $d = \frac{ht}{mx_0}$. A more formal discussion can be given using Wigner function [18].

Clear interference fringes are observed when the BEC is fully separated in our simulation. Therefore, a harmonic potential same as in the last subsection and a stronger Gaussian potential are used. A result is demonstrated in Fig. 2. By fitting the measured fringes periods to time, we calculated the initial space separation distance $x_0 = 50.6\mu$m which agrees roughly with the directly measured initial separation 41.0μm.

IV. SIMULATION OF BEC INTERFEROMETER

In this paper we present a simulation of the experimental work by S. Gupta et al. in 2002 [1], and realized the simulation of a BEC interferometer with a symmetric configuration, as is shown in Fig. 3. After producing the initial state, the atomic wave packet is coherently split into 3 parts, the 0 and $\pm 2\hbar k$ momentum states. After free-evolution of time T, it's coherently reflected and continue to accumulate phase difference. The interference signal is extracted at around $t = 2T$.

At $t = 0$, we use a harmonic potential with frequencies $(20, 50, 50) \times 2\pi$ Hz in the Hamiltonian of our simulation to produce the initial zero momentum ground state. It is calculated using the imaginary time propagation method discussed in Sec. III. The subsequent steps(beam splitting and recombining) involve coherent manipulation of atomic wave packets.

A. Beam Splitting and Reflecting

There are many ways to achieve this manipulation, such as microfabricated structures [3] and simulated Raman transition [19] etc. Here we use standing laser waves as the diffraction gratings of matter wave. Two 1-D counter-propagating traveling waves

$$\begin{aligned}\boldsymbol{E}(z,t) &= \boldsymbol{E}_0 f(t)\sin(kz - \omega t) + \boldsymbol{E}_0 f(t)\sin(kz + \omega t) \\ &= 2\boldsymbol{E}_0 f(t)\sin(kz)\cos(\omega t)\end{aligned} \tag{13}$$

are used to create a near-resonance standing light wave. In our scheme $k = 10^7$ m^{-1} and therefore the lattice period is $\frac{\lambda}{2} = \frac{\pi}{10}\mu$m. In simulation, we replace the

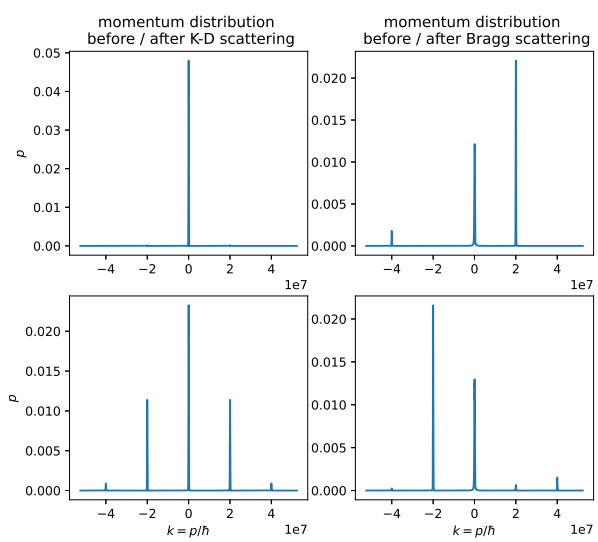

FIG. 4. The effects of the Kapitza-Dirac pulse(left) and Bragg pulse(right), and the vertical axis represents the number density distribution. The left plots show that the initial single momentum state $|g, 0\hbar k\rangle$ becomes a superposition of many momentum states with the intensity of the form $J_n^2(\Omega\tau)$. The right plots show that $|g, 2\hbar k\rangle$ is reflected and becomes $|g, -2\hbar k\rangle$ by the 2nd Bragg π pulse with a nearly 100% reflection rate, which also confirms the correctness of the scattering theory. For clarity the transition from $|g, -2\hbar k\rangle$ to $|g, 2\hbar k\rangle$ is not plotted here.

external potiential V in the established 1-D GPE with $U_0(t)\cos^2(kx)$ corresponding to the light field described by Eq. 13. There are two mechanisms of atom-light interaction that is relied on in our simulation: Bragg scattering and Kapitza-Dirac scattering.

Kapitza-Dirac scattering refers to the conditions where particle motion during interaction can be neglected(the Raman-Nath approximation). In this case, the wave function experiences a space dependent phase modulation

$$|\psi\rangle = |\psi_0\rangle e^{i\Omega\tau\cos(2kz)} \tag{14}$$

when passing through the light field, where $|\Psi_0\rangle$ is the initial wave function, Ω is the two photon Rabi frequency, and τ is the interaction time. When we expand it with the eigenstates of momentum $|g, 2n\hbar k\rangle$ where g means ground state and $2n\hbar k$ is its eigenvalue, we get[20]

$$|\psi\rangle = \sum_{n=-\infty}^{\infty} i^n J_n(\Omega\tau)\,|g, 2n\hbar k\rangle \tag{15}$$

which demonstrates that the scattering momentum spectrum takes the form of Bessel functions of the first kind and $\Omega = 3.7\times10^5\pi\hbar$ in our scheme. In our simulation, we use a temporal square wave of 1μs at $t=0$ to numerically simulate the splitting of the initial wave packet into $|g, 0\hbar k\rangle$ by 43.61%, $|g, \pm2\hbar k\rangle$ each by 24.72% and higher order diffraction terms(neglected in the following derivation and simulation) by 6.68% as shown in the left plots in Fig. 4, consistent with the analytical solution(47.6%, 23.9%, 2.3%).

In Bragg scattering, we consider the light field as two 1-D traveling waves coupled by paired simulated absorption and emission of atoms, rather than a single stationary standing wave. An 1st order Bragg scattering which is used in our scheme involves the process that an atom transfers one photons between the two lasers and gains a momentum of $2\hbar k$. Only energy conserving processes can happen, i.e. an 2nd Bragg scattering can only couple the states $|g, -2\hbar k\rangle$ and $|g, 2\hbar k\rangle$. We give without derivation the transition probability as[20]

$$P_2^B(\tau) = \sin^2\left(\frac{\omega_R^{(4)}}{2}\tau\right), \omega_R^{(4)} = \frac{\Omega^2}{8\omega_{rec}} \tag{16}$$

with which we can control the population on different momentum state, where

$$\omega_{rec} = \frac{\hbar k^2}{2m} \tag{17}$$

. Here, after some time of free propagation, We take $U_0(t)$ as a temporal gaussian function with $\sigma_B = 7.6\mu$s in the external potential at $t=T$ to simulate the 2nd Bragg π pulse's reversing the direction of $|g, -2\hbar k\rangle$ and $|g, 2\hbar k\rangle$ states, as shown in the right plots of Fig. 4.

B. Virtual Contrast Interferometer

The interferometer we demonstrate is based on contrast interferometry, which encodes the photon recoil frequency in the contrast of the interference fringes rather than in their phase[1]. In the simulation of the interferometer, the three paths accumulate phase difference during the $0-2T$ time period and then overlap and interfere at about $t=2T$. The interfering state around $t=2T$ is the sum of three wave packets:

$$|\psi_f\rangle = J_0|\psi_0\rangle + J_1 e^{-i(2kx-4\omega_{rec}t)}|\psi_0\rangle + J_{-1}e^{-i(-2kx-4\omega_{rec}t)}|\psi_0\rangle. \tag{18}$$

The first term interferes with the second and third term respectively, forming two matter wave gratings counter-propagating at a relative velocity $\frac{4\hbar k}{m}$. Their coherent superposition gives the interference pattern of density distribution:

$$\langle\psi_f|\psi_f\rangle = \left[J_0^2 + 4J_1^2\sin^2(2kx)\right]\langle\psi_0|\psi_0\rangle + 4J_0J_1\sin(4\omega_{rec}t)\sin(2kx)\langle\psi_0|\psi_0\rangle \tag{19}$$

The overall density distribution can be regarded as the sum of a temporally dependent wave grating with a spatial frequency $2k$ and a temporally independent one with spatial frequency $4k$. The contrast of the first grating oscillates between 0 and $4J_0J_1$ in a sinusoidal form.

In the original experiment Ref. [1], the matter wave grating is illuminated with a probe light and the contrast signal $S(t)$ is extracted by detecting the intensity of the 1st-order light diffracted by the $2k$ grating (the $4k$ grating does not influence the contrast signal therefore). We

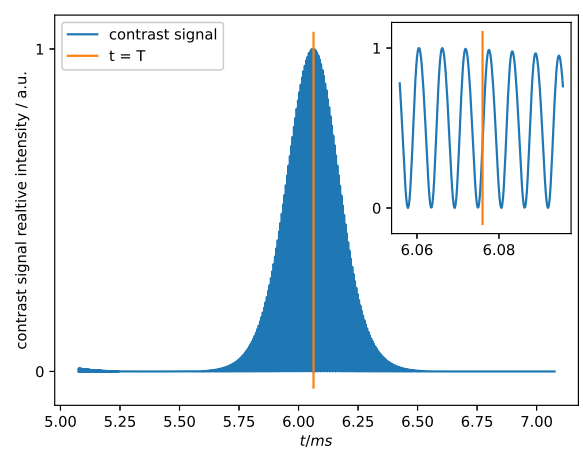

FIG. 5. The extracted contrast signal in a single run. The contrast oscillates in a quadratic-sinusoidal form with an envelope over it. The impact of dephase is slight because in our case the width of wave packets in momentum space are small. The deviation from zero at about $t=5.00$ms is probably the interference of higher order scattered waves due to periodic boundary conditions.

simulate the grating measurement by monitoring the intensity of the $|2\hbar k\rangle$ component in the frequency domain:

$$S(t) = C(t)\sin^2(4\omega_{rec}t) \tag{20}$$

where $C(t)$ is an envelope over the oscillating contrast cause, who has a spread proportional to the coherence time $\frac{1}{k\Delta v}$, where Δv is the atom velocity spread, which again exhibits the advantage of BEC. Fig. 5 shows a typical contrast signal in a single run.

Up to now we are able to simulate the whole process of this BEC contrast interferometry.

V. RESULTS FROM THE INTERFEROMETER

A. Measurement of h

In different runs, the propagation time $2T$ is varied and the corresponding phase $\Phi = 4\omega_{rec}t$ at $t = 2T$ can be measured by fitting the contrast signal $S(t)$ to sinusoidal functions. A linear regression of phase Φ to T gives ω_{rec} and h can be thus calculated from Eq. 17. A typical linear fitting graph is shown in Fig. 6. The measured Planck's constant $\hbar$ is lower than the standard value by 1.418×10^{-4}.

Measurement uncertainty of the slope is comparable to simulation accuracy, which may result from the $\frac{1}{\sqrt{N}}$ decay in slope uncertainty of the Least Square Regression. Keeping in mind the asymmetry of beam splitting mentioned above in our case, the robustness of this contrast interferometry method is clearly shown, and so is in the initial experiment [1].

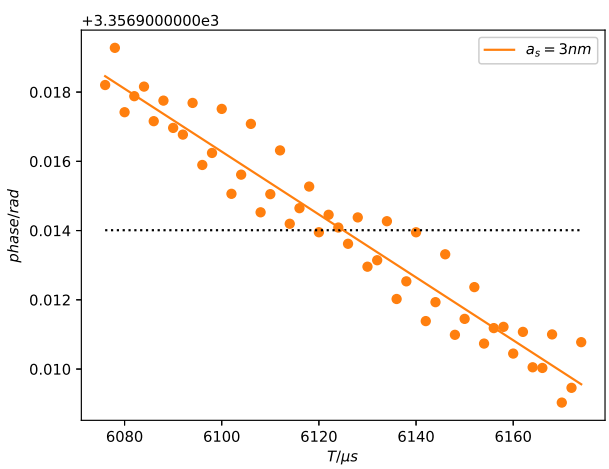

FIG. 6. The fitting data. Measured phase are plotted to T in different runs and the data is subtracted with the standard value. The dotted line represents the slope of standard value. The slope is proportional to Planck's constant h. The contrast signal is measured from $t = 2T - 25\mu s$ to $2T + 25\mu s$ in each run and T is varied in different runs. T starts from 3038 to $3087\mu s$, and 50 runs are executed.

B. The Deviation in h

The deviation of h from the standard value is significant compared to its measurement uncertainty, which is also mentioned in the initial experiment's result [1]. The existence of certain systematic error is therefore indicated. To investigate factors that influences this systematic error, we adjusted the intensity of mean-field interaction by changing the value of scattering length a_s. The results are shown in Fig. 7.

It's surprising that the measured h seem to monotonically decrease from the standard value as the interaction increased. The bias in h was attributed to 'different mean-field shift due to different densities in the 0 momentum path and $\pm|2\hbar k\rangle$ paths' in Ref. [1].

However, we believe the mean field shift will affect no more than signal contrast as long as the wave packet is symmetrically extended during propagation. To demonstrate this, assume Gaussian distribution around intended 0 and $|2\hbar k\rangle$ momentum states. Only interference fringes of frequency $2\hbar k$ is measured, and therefore only components with $2\hbar k$ relative momentum needs to be taken into account. The identical relative momentum make the only effect a shift in the measured phase. The accumulated effect can be represented with an integral:

$$\int_{-\infty}^{\infty} \cos(2kx + \frac{\hbar kT}{m}k')e^{-\frac{(k'-2k)^2}{2\sigma^2}}dk' \tag{21}$$

$$= \sqrt{2\pi}\sigma e^{-\frac{(\sigma\hbar T)^2}{2m^2}}\cos(2kx) \tag{22}$$

Neglecting normalization terms, it's clear that the mean field shift contributes only to the loss of signal intensity. However when k' is not centered around k, a bias pro-

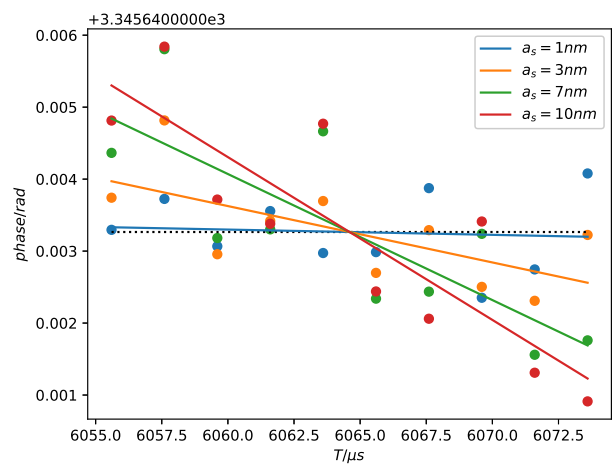

FIG. 7. Measured data and fitting plots in with a_s. The fitting plots are shifted to the center of standard value for clarity. 10 runs are carried out for every scattering length a_s. It can be observed that the fitting slopes deviates more from standard value when a_s increases.

portional to T in the measured phase can be seen:

$$\int_{-\infty}^{\infty} \cos(2kx + \frac{\hbar kT}{m}k')e^{-\frac{(k'-2k-k_\delta)^2}{2\sigma^2}}dk' \tag{23}$$

$$=\sqrt{2\pi}\sigma e^{-\frac{(\sigma\hbar T)^2}{2m^2}} \cos(2kx + \frac{\hbar kT}{m}k_\delta) \tag{24}$$

From the measured momentum distribution shown in Fig. 4, after the π Bragg pulse, $k_\delta \approx -9.06\times10^{-4}2\pi\mathrm{m}^{-1}$. The above calculation yields a 1.134×10^{-4} relative error in h, which aligns with the measured bias. We assume that the scattering process produced this asymmetry. The scattering length a_s decides the initial width of the wave packet in the initial confining potential, so it's presumably the reason for a monotonically decreasing measured h when increasing a_s in Fig. 7.

VI. CONCLUSION

After discussing methods to simulate a BEC and demonstration of BEC's interference features, we constructed a virtual BEC interferometer and achieved a precise measurement of Planck's constant of relative error comparable to simulation resolution, though a systematic error of 1.4×10^{-4} is observed as is in the original experiment. By further simulations and discussions we propose the deviation originates in the beam splitting process. Our virtual interferometer can be used to investigate the various features of BEC and potential factors that can improve the accuracy of the measurement.

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Appendix A: Thomas-Fermi Approximation

Consider a GPE (Eq. 5).

$$i\hbar\frac{\partial\psi}{\partial t} = \left(-\frac{\hbar^2}{2m}\Delta + V + g\|\psi\|^2\right)\psi \tag{A1}$$

If the kinetic energy term $\frac{\hbar^2}{2m}\Delta\psi$ is negligible comparing to the interaction term [10], the wave function can be expressed in terms of the chemical potential, with a form that mirrors the external potential.

$$\psi = \sqrt{\frac{\mu - V(\vec{r})}{g}} \tag{A2}$$

Next we try to fix chemical potential μ. A normalization of wave function instantly yields $\int_D \|\psi\|^2 d^3r = N$, where the integration area depends implicitly on μ.

In our case of a harmonic potential $V_1 = \frac{m}{2}(\omega_x^2x^2 + \omega_y^2y^2 + \omega_z^2z^2)$ plus a Gaussian function oriented in the x direction $V_2 = A\exp(\frac{x^2}{2\sigma^2})$, the boundary is given by

$$\mu = V_1 + V_2 = \frac{m}{2}(\omega_y^2y^2 + \omega_z^2z^2) + U(x) \tag{A3}$$

where $U(x) = \frac{m}{2}\omega_x x^2 + A\exp\frac{x^2}{2\sigma^2}$. The boundary of y and z is a function of x and using the cylindrical symmetry $\omega_y = \omega_z = \omega_\rho$ we get $y^2+z^2 = \rho_{max}^2 = \frac{2\mu-2U(x)}{m\omega_\rho^2}$. The integration of normalization can be simplified as follows:

$$gN = \int_D \mu - V(\vec{r})d\vec{r} \tag{A4}$$

$$= \int_{x_{min}}^{x_{max}} dx \int_0^{2\pi}\int_0^{\rho_{max}(x)} \left(\mu - U(x) - \frac{m}{2}\omega_\rho^2\rho^2\right)\rho d\rho d\phi \tag{A5}$$

As mentioned above, x_{min} and x_{max} depends implicitly on μ. Their relation is given by substituting $y^2 + z^2$ with 0 and x with x_{max} or x_{min} in Eq. A3. In this way, μ can be easily expressed with x_{min} or x_{max} explicitly:

$$\mu = \frac{m\omega_x^2}{2}x_m^2 + A\exp\left(\frac{x_m^2}{2\sigma^2}\right) \tag{A6}$$

Insert Eq. A6 back into Eq. A4, and x_m can be fixed. Eq. A6 then gives μ and wave function ψ can be determined. In our case the chemical potential is approximately 2.8×10^{-30}J. In our simulation of ground states in Fig. 1, the T-F approximation is reliable where the inter-atomic interactions dominate.

Appendix B: Authors' Contributions

The two authors contribute equally to this project, working together on literature research, numerical simulations, discussions, and article writing.

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